



YouTube Trending Videos Web Scrapping

Student Details

Name: Taniya

Roll Number: 8524255

Branch: B.Tech CSE (AI & ML)

College:JMIETI

Company: CodroidHub Private Limited

Instructor:Devashish Kumar

Introduction

This project demonstrates YouTube Trending Videos Web Scrapping using Python and Selenium. Selenium automates the browser to collect video titles and links from YouTube. The extracted data is stored using Pandas and can be exported to a CSV file for further analysis.



Objective

The primary objective of this project is to automate the extraction of YouTube trending video information using Selenium. The project collects video titles and links, organizes the extracted

data into a structured format using Pandas, and exports it to a CSV file. It also provides practical experience in browser automation, dynamic web scraping, and data analysis using Python.

```
In [1]: !pip install selenium webdriver-manager
```

Requirement already satisfied: selenium in c:\users\hp\anaconda3\lib\site-packages (4.45.0)
Requirement already satisfied: webdriver-manager in c:\users\hp\anaconda3\lib\site-packages (4.1.2)
Requirement already satisfied: certifi>=2026.2.25 in c:\users\hp\anaconda3\lib\site-packages (from selenium) (2026.6.6.17)
Requirement already satisfied: trio<1.0,>=0.31.0 in c:\users\hp\anaconda3\lib\site-packages (from selenium) (0.33.0)
Requirement already satisfied: trio-websocket<1.0,>=0.12.2 in c:\users\hp\anaconda3\lib\site-packages (from selenium) (0.12.2)
Requirement already satisfied: typing_extensions<5.0,>=4.15.0 in c:\users\hp\anaconda3\lib\site-packages (from selenium) (4.15.0)
Requirement already satisfied: urllib3<3.0,>=2.6.3 in c:\users\hp\anaconda3\lib\site-packages (from urllib3[socks]<3.0,>=2.6.3->selenium) (2.7.0)
Requirement already satisfied: websocket-client<2.0,>=1.8.0 in c:\users\hp\anaconda3\lib\site-packages (from selenium) (1.8.0)
Requirement already satisfied: attrs>=23.2.0 in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (25.4.0)
Requirement already satisfied: sortedcontainers in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (2.4.0)
Requirement already satisfied: idna in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (3.11)
Requirement already satisfied: outcome in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (1.3.0.post0)
Requirement already satisfied: sniffio>=1.3.0 in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (1.3.0)
Requirement already satisfied: cffi>=1.14 in c:\users\hp\anaconda3\lib\site-packages (from trio<1.0,>=0.31.0->selenium) (2.0.0)
Requirement already satisfied: wsproto>=0.14 in c:\users\hp\anaconda3\lib\site-packages (from trio-websocket<1.0,>=0.12.2->selenium) (1.3.2)
Requirement already satisfied: pysocks!=1.5.7,<2.0,>=1.5.6 in c:\users\hp\anaconda3\lib\site-packages (from urllib3[socks]<3.0,>=2.6.3->selenium) (1.7.1)
Requirement already satisfied: requests in c:\users\hp\appdata\roaming\python\python313\site-packages (from webdriver-manager) (2.34.2)
Requirement already satisfied: python-dotenv in c:\users\hp\anaconda3\lib\site-packages (from webdriver-manager) (1.1.0)
Requirement already satisfied: packaging in c:\users\hp\anaconda3\lib\site-packages (from webdriver-manager) (25.0)
Requirement already satisfied: pycparser in c:\users\hp\anaconda3\lib\site-packages (from cffi>=1.14->trio<1.0,>=0.31.0->selenium) (2.23)
Requirement already satisfied: h11<1,>=0.16.0 in c:\users\hp\anaconda3\lib\site-packages (from wsproto>=0.14->trio-websocket<1.0,>=0.12.2->selenium) (0.16.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\hp\anaconda3\lib\site-packages (from requests->webdriver-manager) (3.4.4)



Imported Libraries

- **selenium.webdriver** – Used to automate the Chrome browser.
- **By** – Helps locate web elements using XPath, ID, Class Name, etc.
- **Service** – Connects Selenium with the ChromeDriver service.
- **ChromeDriverManager** – Automatically downloads and manages the correct ChromeDriver.
- **time** – Provides delays to allow web pages to load properly.
- **pandas** – Stores the extracted data in a DataFrame and exports it to a CSV file.

```
In [2]: from selenium import webdriver
        from selenium.webdriver.common.by import By
        from selenium.webdriver.chrome.service import Service
        from webdriver_manager.chrome import ChromeDriverManager
        import time
```

```
In [3]: def youtube_trending():
        options = webdriver.ChromeOptions()
        options.add_argument("--start-maximized")

        driver = webdriver.Chrome(
            service=Service(ChromeDriverManager().install()),
            options=options
        )

        driver.get("https://www.youtube.com/results?search_query=trending+videos")

        time.sleep(6)

        videos = []
```

```

# Scroll page
for _ in range(5):
    driver.execute_script(
        "window.scrollTo(0, document.documentElement.scrollHeight);"
    )
    time.sleep(2)

# Get all links
elements = driver.find_elements(By.XPATH, "//a[@href]")

print("FOUND:", len(elements))

count = 0

for e in elements:
    try:
        title = e.get_attribute("title")
        link = e.get_attribute("href")

        if title and link and "watch?v=" in link:
            videos.append({
                "Title": title,
                "Link": link
            })
            count += 1

        if count == 20:
            break

    except:
        pass

driver.quit()
return videos

```

```

In [4]: videos = youtube_trending()
        videos

```

FOUND: 259

```
Out[4]: [{ 'Title': 'How to Create Viral Instagram Motu Patlu AI Videos | Trending Motu Patlu AI Video Editing for Ins...',
  'Link': 'https://www.youtube.com/watch?v=0ly8o3w_qbE&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Trending Motu Patlu AI Video Editing | How To Create Instagram Viral Motu Patlu AI Videos',
  'Link': 'https://www.youtube.com/watch?v=JjPZPajgZGs&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Instagram Viral Motu Patlu Wali Ai Video Kaise banaye | Trending Motu Patlu Village Video Editing',
  'Link': 'https://www.youtube.com/watch?v=deuArG_xFTs&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'How to Create Viral Instagram Motu Patlu AI Videos | Trending Motu Patlu Village Video Editing',
  'Link': 'https://www.youtube.com/watch?v=GEstUKBeHKA&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'How to Create Viral Motu Patlu AI Videos on Instagram 🔥 | Trending Motu Patlu City & Village Reel Edit',
  'Link': 'https://www.youtube.com/watch?v=Sh0ExWv-Otc&pp=ygUPdHJlbmRpbmcgdmlkZW9z0gcJCU8LAYcqIYzv'},
  { 'Title': 'How to Make Viral Instagram Motu Patlu AI Videos 🔥 Trending AI Vlog City & Village Reels #motupatlu',
  'Link': 'https://www.youtube.com/watch?v=0SQTy9BpbBI&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'How to Create Viral Instagram Motu Patlu AI Vlog Videos | Trending Motu Patlu Village Video Editing',
  'Link': 'https://www.youtube.com/watch?v=zECj9VWWwf4&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'How to Create Viral Instagram Motu Patlu AI Videos | Trending Motu Patlu AI Video Editing for Ins...',
  'Link': 'https://www.youtube.com/watch?v=0ly8o3w_qbE&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Best TikToks of 2025 So Far 😂 Funniest Viral Videos',
  'Link': 'https://www.youtube.com/watch?v=pGU1W-F7oD0&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Trending Motu Patlu Blogging Video Editing | Instagram Viral Motu Patlu Ai Vlog Video Kaise Banaen',
  'Link': 'https://www.youtube.com/watch?v=w9GnP1m-w6k&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Create AI Videos Free in 2026 | Trending & Viral Ideas',
  'Link': 'https://www.youtube.com/watch?v=afaqQpSAHSg&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'How to Make Trending Motu Patlu AI Videos 🔥 | How to Make AI Motu Patlu Videos of Your Village or...',
  'Link': 'https://www.youtube.com/watch?v=Ae7ozWd68nc&pp=ygUPdHJlbmRpbmcgdmlkZW9z'},
  { 'Title': 'Instagram Trending Motu Patlu AI Video Editing | How to make viral Motu Patlu AI videos',
  'Link': 'https://www.youtube.com/watch?v=6GJ7K6NOotY&pp=ygUPdHJlbmRpbmcgdmlkZW9z'}]
```

```
In [5]: import pandas as pd

df = pd.DataFrame(videos)
df
```

Out[5]:

	Title	Link
0	How to Create Viral Instagram Motu Patlu AI Vi...	https://www.youtube.com/watch?v=0ly8o3w_qbE&pp...
1	Trending Motu Patlu AI Video Editing How To ...	https://www.youtube.com/watch?v=JjPZPajgZGs&pp...
2	Instagram Viral Motu Patlu Wali Ai Video Kaise...	https://www.youtube.com/watch?v=deuArG_xFTs&pp...
3	How to Create Viral Instagram Motu Patlu AI Vi...	https://www.youtube.com/watch?v=GEstUKBeHKA&pp...
4	How to Create Viral Motu Patlu AI Videos on In...	https://www.youtube.com/watch?v=Sh0ExWv-Otc&pp...
5	How to Make Viral Instagram Motu Patlu AI Vide...	https://www.youtube.com/watch?v=0SQTy9BpbBI&pp...
6	How to Create Viral Instagram Motu Patlu AI VI...	https://www.youtube.com/watch?v=zECj9VWWf4&pp...
7	How to Create Viral Instagram Motu Patlu AI Vi...	https://www.youtube.com/watch?v=0ly8o3w_qbE&pp...
8	Best TikToks of 2025 So Far 🤪 Funniest Viral V...	https://www.youtube.com/watch?v=pGU1W-F7oD0&pp...
9	Trending Motu Patlu Blogging Video Editing I...	https://www.youtube.com/watch?v=w9GnP1m-w6k&pp...
10	Create AI Videos Free in 2026 Trending & Vir...	https://www.youtube.com/watch?v=afaqQpSAHSg&pp...
11	How to Make Trending Motu Patlu AI Videos 🔥 ...	https://www.youtube.com/watch?v=Ae7ozWd68nc&pp...
12	Instagram Trending Motu Patlu AI Video Editing...	https://www.youtube.com/watch?v=6GJ7K6NOotY&pp...

```
In [6]: df.to_csv("youtube_trending.csv", index=False)
print("CSV Saved Successfully!")
```

CSV Saved Successfully!



Conclusion

This project successfully demonstrates the process of web scraping YouTube Trending Videos using Python and Selenium. The scraper automatically extracts video titles and their corresponding links from YouTube and stores the data in a structured format. The extracted

information can be further analyzed using Pandas or exported to a CSV file for future use. This project enhances practical knowledge of browser automation, dynamic web scraping, data collection, and Python programming, making it a valuable application of Selenium in real-world scenarios.